

Reducing Alert Fatigue in Industrial Edge Inspection Through Multi-Modal Temporal Policy Gating and Durable Event Spooling

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Abstract—Visual anomaly detectors produce a score per frame, but an inspection line needs decisions that an operator can act on. Thresholding every frame floods the review queue, while camera glare, a blurred lens or a machine in maintenance can each look like a defect. We describe an edge runtime that turns PatchCore scores and three physical sensor channels into operator alerts through exponential smoothing, k -of- N persistence, machine-state gating, cross-modal divergence triage and an incident latch, and that stores every event in a SQLite write-ahead spool until the MQTT broker acknowledges it. We evaluate 8 policy variants on 6 workloads whose visual scores come from PatchCore on 7 MVTec AD categories (1,008 runs of 9,000 frames, 21 paired units per cell). On good parts, single-frame thresholding raised 4,616 false alerts per hour and the full policy 0.0; a sustained defect produced 1.13 alerts instead of 137.64, at a mean delay of 3.7 frames, and a 1–3 frame glare still produced 0.66 alerts on average. Divergence triage cut false line-stop escalations from 117.0 to 28.9 per hour. Against a real Mosquitto broker the spool lost no event under outages of up to 120 s, a broker restart or a publisher crash. On an RTX 4050 Laptop GPU the measured cycle took 24.4 ms on average, but 1.62% of cycles exceeded the 33.3 ms frame budget.

Index Terms—Edge computing, visual anomaly detection, alarm management, sensor fusion, MQTT, store-and-forward.

I. INTRODUCTION

Unsupervised detectors such as PatchCore [1] and PaDiM [2] rank images well on MVTec AD [3], [4], and edge accelerators can run them at line rate [5]. Ranking quality is not the same as a usable alarm stream. A camera at 30 frames per second sees the same defect in dozens of consecutive frames; reflections and dust produce short score spikes; and scores are meaningless while the lens is blurred or the machine is being serviced. Process-industry alarm management has studied the resulting overload for years: ISA-18.2 [6] and EEMUA 191 [7] recommend on-delay timers, deadbands, re-alarm suppression and state-based alarming, and their effect on detection delay has been analyzed [8], [9]. Networking adds a second problem: an alert that is lost during a broker outage is worse than a late one.

This paper applies these ideas to an inspection runtime and measures what each of them contributes. Our contributions are:

- 1) a decision policy that combines smoothing, k -of- N persistence (an on-delay in ISA-18.2 terms), machine-

state gating, divergence-based review routing and an incident latch, with an explicit rule for every degraded input;

- 2) a store-and-forward publisher that deletes a spooled event only after the broker acknowledges it;
- 3) an ablation driven by real PatchCore scores on seven MVTec AD categories, with paired statistics, a parameter sensitivity sweep and an operator-load model;
- 4) measurements of durability against a real broker and of per-stage latency with a real forward pass.

All comparisons are internal ablations. We do not claim novelty for the individual filters, and we do not report factory data.

II. SYSTEM

A. Scoring

PatchCore (ResNet-18 layer2/3 features, 10% greedy k -center coreset [10], 224×224 input) returns the maximum of a smoothed nearest-neighbor distance map. Raw distances differ in scale between products, so we normalize each category with $v = \min(1, 0.5 d/d_{99})$, where d_{99} is the 99th percentile of distances on held-out *training* images; the nominal 99th percentile thus maps to 0.5. Before scoring, an optical check computes the variance of the Laplacian [11] at the model resolution and the mean brightness; frames below a per-category blur threshold (half the 5th percentile on held-out training images) or outside the brightness limits skip the detector. Vibration, temperature and current are standardized against running baselines and fused into $s = 1 - \exp(-\sum_c w_c \max(0, (z_c - 3)/3))$ with weights 0.45/0.25/0.30.

B. Decision policy

Scores are smoothed with $\bar{v}_t = \alpha_v v_t + (1 - \alpha_v) \bar{v}_{t-1}$ and $\bar{s}_t = \alpha_s s_t + (1 - \alpha_s) \bar{s}_{t-1}$ ($\alpha_v = 0.35$, $\alpha_s = 0.25$), each initialized with its first valid sample. A frame that fails the optical check does not update $\bar{v}$, and a reading with a missing channel does not update $\bar{s}$. Visual evidence is *high* when at least $k = 4$ of the last $N = 10$ smoothed scores reach $\tau_{\text{high}} = 0.8$ and *medium* for the same test at $\tau_{\text{med}} = 0.5$; sensor evidence is confirmed by the same k -of- N test at $\tau_{\text{phys}} = 0.7$. The candidate decision is then chosen in order: (1) a machine

in FAULT state yields HIGH; (2) if the camera is degraded, confirmed sensor evidence yields REVIEW, and an optical failure in k of the last N frames yields a REVIEW for camera maintenance; shorter failures are held silently; (3) high visual evidence confirmed by the sensors yields HIGH; (4) high visual evidence without confirmed sensor evidence yields REVIEW, as a sensor fallback if a channel is missing or as a cross-modal discrepancy if $|\bar{v}_t - \bar{s}_t| \geq \tau_{\text{div}} = 0.45$, and HIGH otherwise; (5) confirmed sensor evidence alone, or medium visual evidence, yields REVIEW. In IDLE or MAINTENANCE the candidate is lowered by one level, so review-level evidence is logged but not queued.

C. Incident latch

Each non-normal decision either opens an incident, which emits one alert, or joins the open incident silently; a severity increase inside an incident emits one more alert. The incident closes after $T_{\text{cool}} = 15$ consecutive normal frames (0.5 s). Only alerts enter the operator queue; every decision is still written to the audit log with its incident identifier. We fixed all parameters before running the experiments; the artifact lists each one with its meaning.

D. Durable delivery

The runtime first inserts every QoS 1 event into a SQLite pool in WAL mode with `synchronous=NORMAL` [12], keyed by a unique event identifier. One drain thread publishes spooled events in insertion order over MQTT 3.1.1 [13] and deletes a row only in the `on_publish` callback, i.e. after the broker’s PUBACK. Delivery is at-least-once: a crash between PUBACK and deletion re-sends the event, and the subscriber and the audit log discard repeated identifiers. At capacity, the pool evicts its oldest rows and counts them. The guarantee covers broker outages and process crashes, not power loss, for which `synchronous=NORMAL` may lose the last transactions.

III. EVALUATION METHOD

A. Score banks

For each of the 7 MVTec AD categories, we fitted PatchCore on a random 80% of the good training images and used the remaining 20% to calibrate d_{99} and the blur threshold (the companion carbon study [14] uses a different 60/20/20 split, so its thresholds are not interchangeable with ours). We scored every test image, and every good test image three more times with a synthetic specular highlight (a blurred saturated ellipse). Image AUROC ranged from 0.925 to 1.000. On 5 of the 7 categories the median normalized defect score lay below τ_{high} , and the median glare score was at least 0.57 in every category, i.e. PatchCore responds to glare much as it does to defects.

B. Workloads

Each workload lasts 9,000 frames (5 min at 30 Hz) and places randomized events without overlap (sensor drift and dropout windows may overlap other events): good parts only; 40 glare bursts of 1–3 frames plus short blur; eight defects of 2–8 s;

defects with a mechanical fault, mechanical faults alone and 10 glare events of up to 2 s; defects around blur, occlusion, current dropout and a temperature drift ramp; and defects between idle and maintenance windows in which glare and mechanical load occur. Per frame, the visual score is drawn from the bank pool matching the timeline (good, defect or glare), the optical check runs on a real held-out frame that is blurred or darkened when the timeline says so, and the sensor simulator follows the timeline’s faults. Defects and mechanical faults while running are ground truth; everything else is not. An experimental unit is a (category, seed) pair; all policies see identical inputs per unit, giving 21 paired units per cell.

C. Policy variants

BASILINE thresholds every raw visual score ($v \geq \tau_{\text{high}}$ gives HIGH, $v \geq \tau_{\text{med}}$ REVIEW); EMA_ONLY applies the same thresholds to $\bar{v}$; and EMA_KOFN adds the k -of- N test. These three are vision-only. FULL_POLICY is the complete policy of Section II, and NO_COOLDOWN, NO_FUSION, NO_DIVERGENCE and NO_STATE_GATING each remove one component from it (without fusion, divergence triage is also off). The companion study [14] calls BASILINE, EMA_ONLY, EMA_KOFN and FULL_POLICY B0–B3.

D. Metrics and statistics

An *alert* is a decision that opens or escalates an incident (for policies without the latch, every non-normal frame). Alerts within an episode or up to 15 frames after it belong to that episode; other alerts are false (*critical* if at HIGH). Recall is the fraction of episodes with a non-normal decision within 15 frames of onset, and delay is measured to that decision. False alerts are normalized by the time outside episodes, and glare responses by the number of bursts. We report means over units with 95% percentile-bootstrap intervals ($B = 2,000$) in the artifact, and compare each policy with the baseline by Wilcoxon signed-rank tests with Holm correction.

IV. RESULTS

A. Policy ablation

Table I summarizes the ablation. With good parts only, single-frame thresholding produced 4,616 false alerts per hour, because normalized scores of good test images regularly cross 0.5; smoothing alone removed almost all of them, and the full policy left 0.0 per hour (no false alert in any unit). Glare is harder: the smoothed score of a 3-frame burst stays above τ_{med} long enough to pass the k -of- N test, so persistence alone *increased* alerts per burst (5.66 vs. 2.63), and only the incident latch brought them to 0.66. The latch also turned a sustained defect from 137.64 alerts into 1.13, reducing alerts per hour in that workload by 99.4% (17,152 to 109), at a mean delay of 3.7 frames (122 ms). Without sensor fusion, recall on the multi-modal workload fell from 1.00 to 0.62, since mechanical faults without a visible defect were never seen. Divergence triage lowered false HIGH escalations from 117.0 to 28.9 per hour by sending visual evidence that disagrees with the sensor score, including 2-second glare, to review. State gating lowered

TABLE I

ABLATION OVER 21 PAIRED UNITS PER CELL (MEANS). FA/H: FALSE ALERTS PER HOUR OF NON-DEFECT TIME. ALERTS/BURST: ALERTS PER GLARE BURST. ALERTS/EP.: ALERTS PER SUSTAINED-DEFECT EPISODE (1 IS IDEAL). CRIT. FA/H: FALSE ALERTS AT HIGH SEVERITY. CONFIDENCE INTERVALS AND ALL REMAINING METRICS ARE IN RESULTS/ABLATION/ABLATION_SUMMARY.JSON.

Policy	Nominal	Glare	Sustained defects		Multi-modal faults		Degraded	States
	FA/h	alerts/burst	alerts/ep.	delay (fr.)	recall	crit. FA/h	FA/h	FA/h
BASELINE	4,616	2.63	137.64	0.1	0.78	3,342	4,253	7,299
EMA_ONLY	40.6	4.27	149.21	0.6	0.62	3,229	37.6	3,179
EMA_KOFN	0.0	5.66	153.95	3.7	0.62	3,649	0.0	3,389
NO_COOLDOWN	0.0	5.66	153.95	3.7	1.00	181.2	18,526	24.2
NO_FUSION	0.0	0.66	1.38	3.7	0.62	117.7	107.5	66.2
NO_DIVERGENCE	0.0	0.66	1.38	3.7	1.00	117.0	120.9	66.2
NO_STATE_GATING	0.0	0.66	1.13	3.7	1.00	28.9	120.9	135.9
FULL_POLICY	0.0	0.66	1.13	3.7	1.00	28.9	120.9	15.3

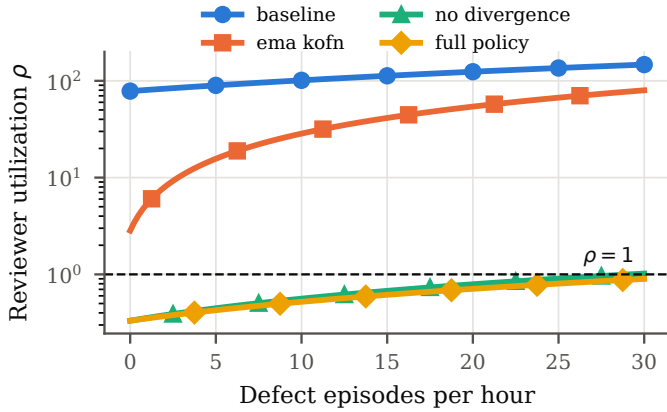


Fig. 1. Modeled utilization of one reviewer (assumed $\mu = 60$ reviews/h) as a function of the defect-episode rate, using the measured alerts per episode, alerts per glare burst (assuming 30 bursts/h) and nominal false-alert rate of each policy. Values above $\rho = 1$ mean an unbounded backlog under an $M/M/1$ model.

false alerts around maintenance from 135.9 to 15.3 per hour. In the degraded-input workload 147 of the 324 alerts of the full policy were camera-maintenance requests for sustained blur, and 42 were caused by the temperature drift, which the policy cannot tell apart from real heating. Across workloads, the full policy differed from the baseline in alerts per hour with Holm-adjusted $p \leq 6.4 \times 10^{-4}$.

Persistence also sets a floor on the length of a defect the policy can catch. In runs with 20 visual defects of one fixed length each (same units, nominal sensors), the full policy found 0.10 of 1-frame, 0.43 of 2-frame and 0.70 of 3-frame defects, against 0.96 of 1-frame defects for the baseline. Smoothing carries evidence past the end of a defect, so some defects shorter than $k = 4$ frames are still caught, but the full policy reached a mean recall of 0.95 only from 8 frames (267 ms) on. The recalls in Table I therefore apply to defects at least this long; a part that stays in view for fewer frames can pass unflagged.

B. Operator load

The workloads compress many events into five minutes, so their hourly alert rates overstate a real line. Fig. 1 instead combines the per-event measurements with two assumed rates that we did not measure: 30 glare bursts per hour and a service rate of $\mu = 60$ reviews per hour, i.e. one minute per alert (the companion study [14] varies the review time from 30 to 90 s). Under an $M/M/1$ review model [15], one reviewer is overloaded by the baseline even without defects, and by EMA_KOFN because of glare alone. The full policy stays below $\rho = 1$ up to about 35 defect episodes per hour at these rates. Queue load is not the only cost of false alerts: operators tend to match their response rate to an alarm’s reliability [16], and the resulting “cry wolf” effect has been examined in operational alerting [17]. At one defect episode per hour, only 5% of the full policy’s alerts would belong to a defect, most of the rest coming from glare; glare suppression therefore matters most on slow lines, where defects are rare. Alerts from the policy are burstier than Poisson arrivals, so real waiting times would exceed the model.

C. Sensitivity

Fig. 2 varies one parameter at a time. Raising k from 2 to 6 lowered false critical escalations from 58.6 to 6.7 per hour and glare false alerts by more than half, at the cost of delay, which grew from 1.6 to 5.7 frames. A looser divergence threshold or faster smoothing let more glare reach HIGH. Longer incidents (T_{cool} up to 90 frames) lowered glare alerts by about a quarter but changed critical escalations little. Recall on the multi-modal workload never fell below 1.00. The defaults were not tuned on these results, and several neighbors trade delay for fewer alerts.

D. Latency

Table II times every stage with a real PatchCore forward pass. On the RTX 4050 Laptop GPU the mean cycle took 24.4 ms (p95 27.2 ms, p99 36.1 ms), dominated by PNG decoding and the 13.6 ms vision stage, while the policy needed 0.15 ms at p99. Even so, 1.62% of cycles exceeded 33.3 ms (worst 55.6 ms). Decoding and scoring at their 95th percentiles already use most

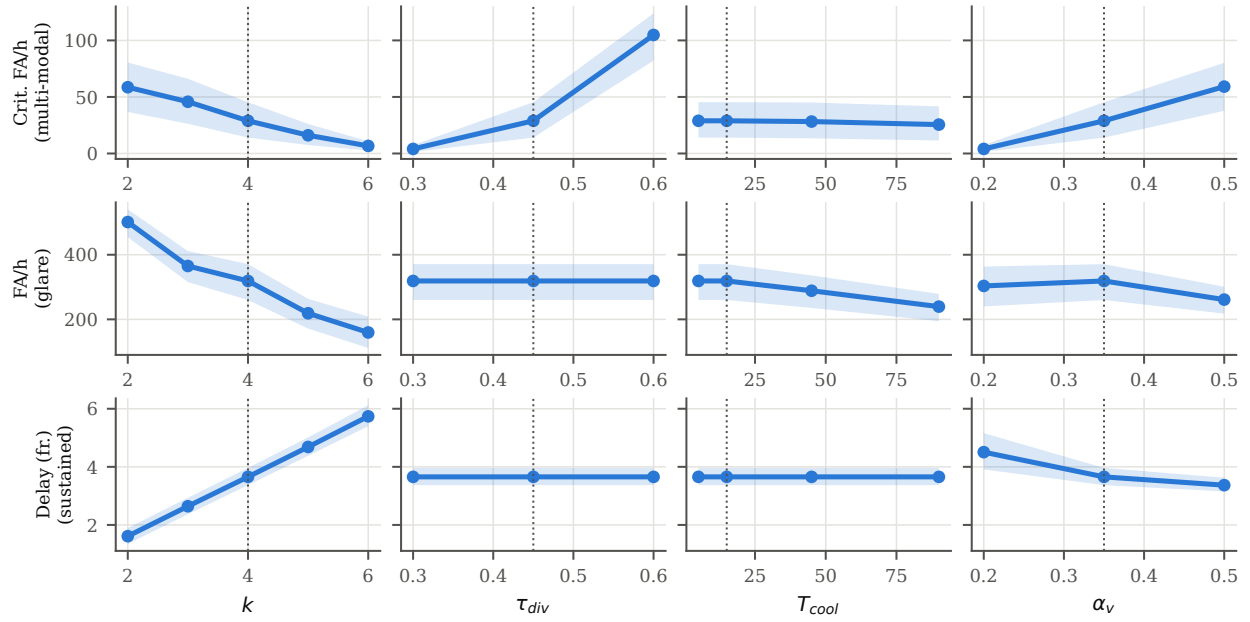


Fig. 2. One-at-a-time sensitivity of the full policy (mean and 95% bootstrap band over 21 units; dotted line: fixed default). Rows: false critical alerts per hour (multi-modal workload), false alerts per hour (glare workload), mean detection delay (sustained defects).

TABLE II
PER-CYCLE LATENCY (5,000 GPU CYCLES, 500 CPU CYCLES; 700×700 PNG FRAMES; MEMORY BANK 13,798 PATCHES).

Stage (ms)	GPU			CPU	
	mean	p99	max	mean	p99
PNG decode	10.06	13.05	23.8	10.99	14.13
Optical check + PatchCore	13.59	16.53	31.2	51.41	73.07
Sensor step	0.11	0.22	0.4	0.12	0.30
Policy	0.06	0.15	0.4	0.07	0.18
Spool insert	0.32	0.55	27.0	0.31	0.59
Audit insert	0.22	4.84	28.5	0.23	2.45
End to end	24.36	36.06	55.6	63.12	88.76

of the budget, and persistence added occasional spikes of up to 27 ms for a spool insert and 28 ms for an audit insert, which suggests moving it off the inspection thread. On the CPU alone the cycle took 63.1 ms and missed the budget in 100% of cycles. On embedded hardware, the companion study [14] compiled the same ResNet-18 PatchCore scoring path to TensorRT 10.16 in FP16 on a Jetson Orin Nano (L4T R39, 15 W mode): paced at 30FPS, it took a median of 8.2 ms per frame (p95 8.6 ms) including host-device copies, reached 243FPS unpaced, and the whole module drew 3.98 W (0.85 W above idle), with alert decisions identical to FP32.

E. Durable delivery

Table III reports six fault cases. Link losses of up to 120 s (3,751 events), a broker restart and a SIGKILL of the publisher during draining lost no event, delivered none out of order, and drained within 2.0 s after recovery. When a 60 s outage exceeded a 1,000-record spool, the spool evicted 802 events and 801 were missing, so every loss was accounted for (0

TABLE III
DELIVERY THROUGH A REAL MOSQUITTO 2.0.18 BROKER AT 30 EVENTS/S (QoS 1). MISS.: EVENTS NEVER RECEIVED; EVICT.: REMOVED BY THE SPOOL CAPACITY; DUP.: REPEATED DELIVERIES; ORDER: FIRST DELIVERIES OUT OF SEQUENCE.

Fault	Events	Miss.	Evict.	Dup.	Order	Peak	Drain (s)
Link loss 30 s	1,051	0	0	0	0	902	0.50
Link loss 60 s	1,951	0	0	0	0	1,802	1.02
Link loss 120 s	3,751	0	0	0	0	3,602	2.03
Broker restart 30 s	1,051	0	0	0	0	901	1.58
Publisher SIGKILL	600	0	0	0	0	440	1.43
Overflow (cap. 1,000)	1,951	801	802	1	0	1,000	0.54

TABLE IV
REPLAY OF TWO SYNTHETIC RUN-TO-FAILURE TRACES (600 FRAMES, SHAPE INSPIRED BY THE IMS BEARING [18] AND C-MAPSS [19] DATA BUT NOT DERIVED FROM THEM), WITH GOOD-PART VISUAL SCORES, AVERAGED OVER THE SEVEN CATEGORIES.

Trace	Policy	pre-fault FA/h	recall	delay (fr.)
Bearing proxy	BASELINE	4,243	0.43	8.3
	NO_FUSION	0	0.00	–
	FULL_POLICY	300	1.00	0.0
Thermal-creep proxy	BASELINE	4,208	0.57	3.5
	NO_FUSION	0	0.00	–
	FULL_POLICY	327	1.00	0.0

unexplained); the other 1 evicted event had already been sent before its row was evicted and still arrived. The crash run happened to produce no duplicates; they are possible by design and are removed by identifier.

F. Synthetic sensor traces

Table IV checks the sensor path in isolation. A vision-only policy cannot see these faults, so the baseline's few detections are chance crossings of nominal scores. The full policy flagged both faults at onset (recall 1.00); its pre-fault alerts are review requests during the incipient drift that precedes the labeled onset, which a user may regard as early warnings rather than errors.

V. LIMITATIONS

All visual scores come from 7 curated MVTec AD categories photographed under controlled conditions. With the same ResNet-18 PatchCore at the 99th-percentile threshold, the companion study [14] measured a median per-image recall of 0.94 on MVTec AD but 0.57 on the 12 VisA categories, whose images are closer to production, and 0.72 on VisA at 448×448 input. The policy can delay or merge detections but not create them, so such detector losses carry over to the alert stream. The sensor channels are simulated and the traces are synthetic. Within an episode, visual scores are sampled independently, whereas consecutive video frames are correlated; the glare is synthetic; and event densities are much higher than on a real line, which is why Section IV-B models load per event. The blur check misfired on real defects: in metal_nut it flagged 19.4% of defective test images (flip) as blurred, which would route those parts to a camera-maintenance review instead of a defect review, while no good test image was flagged. Our latency was measured on one laptop GPU without camera acquisition or network transport. The queue model assumes Poisson arrivals and one reviewer, and its glare and review rates are assumptions. Finally, we compare only with ablations of our own policy, not with other alarm-rationalization or fusion methods.

VI. CONCLUSION

Temporal gating is useful only in combination: persistence removes most single-frame noise but, without an incident latch, multiplies alerts for anything that lasts a few frames; sensor fusion is needed to see faults the camera cannot; and divergence triage and state gating decide mostly *how* evidence reaches the operator. Acknowledgment-driven spooling kept every event through the outages we tested. The remaining gaps are the latency tail, glare that lasts several frames, defects visible in only a few frames, and a blur check that confuses some defects with defocus. Code, configuration, scores and every generated table and figure are available at https://github.com/sengarutk/edge-quality-intelligence;scripts/reproduce_all_paper_results.sh regenerates them.

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